BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Male partner unknown HIV status' as a risk factor for HIV incidence and clinical outcomes in PMTCT programmes in 21 WHO priority countries, a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-057190
Article Type:	Protocol
Date Submitted by the Author:	08-Sep-2021
Complete List of Authors:	Mmotsa, Tshiamo; South African Medical Research Council, HIV Prevention Research Unit Ngandu, Nobubelo; South African Medical Research Council, HIV Prevention Research Unit Adetokunboh, Olatunji; Stellenbosch University, DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA) Nyasulu, P; Stellenbosch University, Division of Epidemiology and Biostatistics
Keywords:	EPIDEMIOLOGY, HIV & AIDS < INFECTIOUS DISEASES, Community child health < PAEDIATRICS

SCHOLARONE™
Manuscripts

Male partner unknown HIV status' as a risk factor for HIV incidence and clinical outcomes in PMTCT programmes in 21 WHO priority countries, a systematic review protocol

Tshiamo Moshading Mmotsa: HIV Prevention Research Unit, South African Medical Research Council, Cape Town, South Africa

Nobubelo Kwanele Ngandu (corresponding author): HIV Prevention Research Unit, South African Medical Research Council, P. O. Box 19070, Tygerberg 7505, Cape Town, South Africa. Email: Nobubelo.Ngandu@mrc.ac.za

Olatunji O Adetokunboh: DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, Stellenbosch, South Africa. Division of Epidemiology and Biostatistics, Department of Global Health, Stellenbosch University, Cape Town, South Africa

Peter Nyasulu: Division of Epidemiology and Biostatistics, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa. Division of Epidemiology and Biostatistics, School of Public Health, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa

Word count = 3102

For peer review only

Abstract

Introduction Research has shown an association between increased disclosure of Human Immunodeficiency Virus (HIV) status by pregnant and breastfeeding women and improved clinical health and that of their infant. Increasing awareness about their male partner's HIV status will no doubt lead to even better outcomes at the population level. Male partner involvement is important for improving outcomes of prevention of mother-to-child transmission of HIV (MTCT) as it improves social support and commitment from both parents of the baby to ensure sustained good health. Although lack of knowledge of the HIV status of a male partner is of great concern, limited research has been done to determine whether it remains one of the barriers to reaching the proposed goals of eliminating MTCT in pregnant or postpartum women. Our aim is to determine if lack of knowledge of a male partner's HIV status is a significant risk factor for HIV incidence and poor HIV clinical outcomes among pregnant women and postpartum women and their infants.

Methods and analysis A systematic review and meta-analysis of experimental and observational studies will be conducted. The review will focus on knowledge of male partner's HIV status in the 21 priority countries most affected by HIV in Africa. We will search electronic databases such as PubMed/MEDLINE, Scopus, Web of Science and Cochrane library, Science Direct, CINAHL, LILACS and SciELO databases from January 2011 to December 2021. We will also search the Pan African and WHO clinical trial registries and conference archives. We will conduct a quality assessment of eligible studies and evaluate the heterogeneity of the pooled studies using the I^2 statistic. The statistical analysis will be performed using STATA statistical software version 16.

Ethics and dissemination The study will use publicly available data and ethics exemption has been obtained from Human Research Ethics Committees, Faculty of Medicine & Health Sciences, Stellenbosch University. The protocol was registered on Prospective Register of Systematic Reviews, registration number CRD42021247686, in May 2021. Findings of this systematic review will be disseminated in peer-review journals including various media platforms i.e., webinars, symposia, conferences or congresses.

Keywords male-partner, HIV disclosure, mother-to-child transmission of HIV, pregnant and postpartum, breastfeeding women, systematic review

For peer review only

Prospective Register of Systematic Reviews, registration number CRD42021247686.

This is an open access article distributed in accordance with the Creative Commons Attribution Non-Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

Strengths and limitations of this study

- This study protocol outlines a path that will minimise duplication of data and provides clear and transparent methods and procedures to be used.
- This study will provide quality evidence through a rigorous methodological process that will be used to inform public health decisions in the prevention of mother to child transmission of HIV.
- This systematic review will arguably be the first to explore the risk of Male partner unknown HIV status on HIV incidence and clinical outcomes in PMTCT programmes in 21 WHO priority countries.
- There is generally scarce randomized and non-randomised trials done assessing the impact of male involvement in MTCT of HIV programs as well as studies specifically focussed at the impact of lack of knowledge of male HIV status on maternal and child health outcomes.
- There is potential limitation arising from publication bias of MTCT intervention studies and the challenges with methodological quality of studies found in the grey literature around this subject.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Introduction

Human Immunodeficiency Virus (HIV) remains a global pandemic and in 2019, 38 million people were living with HIV [1]. More than seven percent of those living with HIV were children and sub-Saharan Africa accounts for 88% of these infections [2]. For almost two decades, it has been a top priority to eliminate HIV incidence amongst pregnant and postpartum women and their children. The Global Plan towards the elimination of new HIV infections among children by 2015 and keeping their mothers alive was released in 2011 with a goal to "reduce the number of new infections among children by 90%" and "reduce the number of Acquired Immunodeficiency Syndrome (AIDS)-related maternal deaths by 50%"[3]. In 2010, the World Health Organization (WHO) recommended provision of lifelong antiretroviral therapy (ART) to pregnant women regardless of CD4 count, known as the Prevention of Mother-to-Child Transmission of HIV (PMTCT) Option B+ policy [4]. This approach would simplify HIV care and improve health outcomes particularly in the 21 countries which had limited access to CD4 testing services and partner testing, high fertility rates, prolonged breastfeeding periods and a high generalized HIV epidemic [4].

Despite lifelong ART under the Option B+ regimen, the PMTCT programmes have been slower than anticipated in eliminating HIV incidence in mothers and children due to several factors. Various studies have highlighted risk factors for Mother-to-Child Transmission of HIV (MTCT) to include a high antenatal HIV prevalence (during pregnancy) which has remained around 30% during the past decade and new maternal infections with delayed diagnosis during pregnancy and breastfeeding [5]. A nationally representative study in Malawi, assessing the option B+ programme found that the uptake of ART and the time of initiation are very important in the PMTCT cascade as they affect the effectiveness of the programme[6]. Similarly, another study found that the timing of ART initiation significantly plays a role in improving the role of adherence in the PMTCT programme [7]. The timing and adherence of ART has been reported to be important for effective outcomes, for example the HPTN 052 study among sero-discordant couples in various countries including some countries in Africa found that early ART initiation reduced the transmission of HIV by up to 96%[8]. Therefore, delayed ART initiation and non-adherence is likely to increase maternal viral load which results in higher risk for MTCT [9].

A study in a high prevalence setting in Johannesburg, South Africa, found that some of the risk factors of MTCT and poor ART outcomes (particularly high maternal viral load) as a result of not being initiated on treatment timeously, included lack of antenatal clinic visits during the first trimester of pregnancy as well as unplanned pregnancies [10]. These findings are similar

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

to those of a study conducted in Cape Town by Brittain et al.[11] which found that when pregnancy was unplanned then the women would have an elevated viral load. In addition, studies have shown that younger women and adolescents are also more likely to have poor ART outcomes and to vertically transmit HIV to their infants [12, 13]. Fear of disclosure is another risk factor as it may lead to intimate partner violence. Reduced partner support for PMTCT or ART initiation,[14] multiple pregnancies as well as lower level of education attainment in women were also factors that influenced ART uptake or adherence in PMTCT programmes [5, 7, 10, 15].

Other risk factors for not achieving PMTCT targets include women not knowing their own HIV status or their male partner's HIV status, limited involvement of the male partner in reproductive health services and low partner testing,[10, 13, 16, 17] thus resulting in delayed ART initiation [16]. Since the end of 2014, pregnant women HIV disclosure rates have been increasing. Results from a meta-analysis found that more than 67% of pregnant women in sub-Saharan Africa had disclosed their HIV status [5]. Another study found that 61% of women had disclosed their HIV status to their partner by the end of the pregnancy [11]. An increase in disclosure rates was also evident in subsequent studies conducted after 2015, with disclosure rates of 80%, 85%, 94.9% and 99% in Kenya, Nigeria, Malawi and Uganda, respectively [6, 19-21]. The women in these studies disclosed mostly to their partners and not their family members [5, 6, 19] and in one study disclosure rates were higher in younger women than in older women [20]. According to Bucagu et al.,[22] HIV status disclosure is more frequent in women than in males and disclosure is generally higher in steady partners than in casual partners.

Non-disclosure of both male partners and pregnant or postpartum women was a strong predictor of suboptimal ART adherence and MTCT [14]. Similarly, a systematic review on factors that influence maternal and infant ART adherence found that HIV status non-disclosure in HIV positive pregnant and postpartum women to their male partners or family was one of the factors resulting in sub-optimal ART adherence [15]. Non-adherence to ART and poor ART outcomes were significantly associated with increased proportions of non-disclosure among pregnant and postpartum women [7, 22, 23]. Prior to PMTCT programmes in most countries that encouraged pregnant and postpartum women to disclose their HIV status, non-disclosure made it difficult for women to comply with attendance or participation in PMTCT programmes resulting in greater rates of loss-to-follow-up and poorer ART outcomes [21]. This shows that disclosure, especially to male partners, is important for improved ART outcomes thus prevention of HIV mortality and reducing HIV incidence in infants as well as keeping the mothers healthy.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Although pregnant and breastfeeding women have generally improved in disclosing their own HIV status [5, 6, 19-21], it remains unclear whether knowing their male partner's HIV status has also improved over the years. The uptake of HIV Testing Services (HTS) among men is generally low, and research has shown that male partners of pregnant or postpartum women have no concerns when it comes to knowing their HIV status or with HIV prevention [24]. In a 2018 community-based study conducted in the Delanta District in Ethiopia aiming to assess the involvement of men in HIV testing and counselling, it was found that men also had fears when it comes to testing for HIV, only 57% males were involved in HTS [25]. A study in Kenya found that although the rates of testing and disclosure were low, a home-visit intervention increased HTS uptake in men [26]. In a facility-based study conducted between August 2011 and March 2012 across all nine provinces in South Africa, the prevalence of not knowing a male partner's HIV status amongst pregnant and breastfeeding women, was reported to be as high as 34%[27]. In sub-Saharan Africa, condom use among high-risk populations including young women of reproductive age has declined since 2010 [28]. This is alarming as this may include pregnant women from high-risk settings. Qualitative studies in South Africa, one in Cape Town and another in Mpumalanga found that there was a reduction in consistent condom use once women fell pregnant [29, 30]. Similarly, De Schacht et al.,[13] found that there was reduction in condom use among pregnant women, with 39 women with inconsistent condom use sero-converting compared to zero among those who used a condom always.

Research has shown that there is an association between increased disclosure of pregnant and breastfeeding women's HIV status and improvement in their HIV clinical health and that of their infants [22]. Increasing awareness about their male partner's HIV status will no doubt lead to even better outcomes at the population level. Figure 1 shows a summary of reported and potential (red arrows) role of male-partner HIV disclosure within the PMTCT causal pathways. A few studies have reported improved maternal and infant HIV clinical outcomes among women who had disclosed their HIV status and whose male partners disclosed their HIV status [14, 21, 22]. Male partner involvement is important for improving outcomes of PMTCT programmes as it improves social support and commitment from both parents to ensure the safety of the infant. Although not knowing a male partner's HIV status is a potential concern, limited research has been done to determine whether it remains one of the barriers to reaching the proposed goals to eliminate MTCT and HIV in pregnant or postpartum women. This systematic review and meta-analysis aims to determine whether not knowing a male partner's HIV status is a significant risk factor for HIV incidence and poor HIV clinical outcomes among pregnant and postpartum women and their infants in the 21 WHO priority countries for eliminating MTCT.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Methods and analysis

The protocol has been developed in accordance with the 2015 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P) (Supplementary file 1) [30]. This systematic review is registered with the United Kingdom's National Institute of Health Research (NIHR) Prospective Register of Systematic Reviews (PROSPERO) registration number CRD42021247686.

Inclusion criteria

All studies that report the knowledge of the HIV status of the male partner as an exposure or predictor factor for HIV incidence, viral load or CD4 count or WHO HIV clinical disease stage during pregnancy or postpartum period among women or their infants aged 0-24 months from the 21 priority countries most affected by HIV in Africa (Table 1), will be reviewed. The studies should also have been conducted after the adoption of the PMTCT Option B+ regimen in these listed countries (from the year 2011). We will include English language or articles translated into the English language. Studies reporting measures of association, i.e., experimental (randomized control trials and non-randomized trials) and observational studies (case-control, cohort, and cross-sectional studies) will all be included.

Exclusion criteria

Publications not from the priority countries listed in Table 1 or reporting outcomes from the option A or option B policies, or outcomes that are not HIV incidence, viral load, CD4 count, and WHO HIV clinical disease staging will not be included. Qualitative interviews will not be included.

Table 1: WHO/UNAIDS global priority countries and adoption of PMTCT Option B+ regimen.

Countries Included	
Angola	Mozambique
Botswana	Namibia
Burundi	Nigeria
Cameroon	Tanzania
Chad	The Democratic Republic of the Congo
Côte d'Ivoire	South Africa
Ethiopia	Swaziland

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ghana	Uganda
Kenya	Zambia
Lesotho	Zimbabwe
Malawi	

Information sources and search strategy

A comprehensive search will be conducted in the following databases; PubMed (via MEDLINE), the Cochrane Library, Scopus, Web of Science, Science Direct, CINAHL, LILACS and SciELO. Additional searches will be conducted from the Pan African Clinical Trials Registry (PACTR) (https://pactr.samrc.ac.za/), International Clinical Trials Registry Platform (ICTRP) (https://www.who.int/clinical-trials-registry-platform) as well as published conference abstracts from the International AIDS Society (IAS) Conference and the International Conference on AIDS and STIs in Africa. Reference lists of included studies will also be reviewed to identify any eligible studies that could be added.

The systematic search will be done using various combinations of synonyms of the key search terms that define the Population and the overarching Outcome from the Participant/Exposure/Comparison/Outcome/Time-period/Setting (PECOTS) elements of the research question (Table 2). Search terms will include: 'HIV' or 'AIDS' or 'MTCT' for the Outcome and 'Pregnant' or 'Breastfeeding' or 'Postpartum women' for the Population. A pilot exercise will be conducted to identify other synonyms and search terms/phrases needed to develop an exhaustive search strategy. All studies found will be uploaded into RAYYAN—a systematic review managing software for further processing [31] and duplicates articles will be removed**.**

Table 2: The PECOTS framework

P- Participants	HIV positive pregnant and breastfeeding women, HIV negative pregnant and breastfeeding women, Infants born to HIV positive women
E- Exposure	Lack of knowledge of male partner's HIV status
C- Comparison	Knowledge of male partner's HIV status
O- Outcome	Unsuppressed viral load, HIV incidence, Advanced HIV disease (defined using CD4 count or WHO clinical staging)
T- Time period	Post- adoption of the PMTCT Option B+ regimen in that country
S- Setting	21 WHO priority countries

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Selection process

The selection of studies will be done in duplicate by two reviewers (TM, NN) and discrepancies will be resolved by discussing with a third reviewer (PN, OA). The screening and selection of the manuscripts will be done in several steps and informed by the inclusion criteria of the PECOTS.

Proposed steps to guide the screening and selection process of manuscripts that will be uploaded on to the RAYYAN system are presented in Figure 2 in the following order: (i) titles and abstracts will be screened for the research question's Population and Outcomes, full text articles from this step will be downloaded; articles with the following characteristics will then be selected: (ii) meets the study Setting (WHO priority country) and the Population and Outcomes of the inclusion criteria; (iii) publishing data from the defined Time period (during Option B+ policy); (iv) meet the study design as per inclusion criteria (experimental and observational studies); and (v) have included the Exposure of interest in the data as a predictor/exposure variable.

Data Extraction

Data from the selected full-text manuscripts will be extracted independently by the two reviewers (TM, NN) using a standardised piloted extraction form. Any discrepancies will be resolved by an agreement between the two reviewers or the third if there is no resolution (OA, PN). In cases where additional information is required such as clarifying on information, missing information or regarding relevant data that may not have been included in the manuscript, the authors will contact the corresponding author of the primary studies to access the information.

The following data parameters will be extracted; author, title of the study, study setting, study design, study sample and population, year of study, measure of association or effect between the exposure of interest and HIV incidence/clinical disease.

Quality of manuscripts

Study quality and risk of bias will be assessed by evaluating the study design, data collection as well as the data analysis methods. The quality assessment will be double-checked by the review team. The risk of bias of randomised studies will be assessed using the Cochrane Risk of Bias Tool [32]. The Risk of Bias for Non-Randomised Studies- of Interventions (ROBINS-I) Tool [33] will be used for non-randomised trials while the Quality in Prognostic Studies (QUIPS) Tool [34] will be used for observational studies. The summary of characteristics of each study will be presented in a table format.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Synthesis of results

A meta-analysis will be performed if the selected manuscripts are homogenous in terms of methodology and outcomes. Data that is sufficiently similar will be pooled together using the inverse variance approach to accommodate crude and adjusted measures of association and effect, where possible. Additionally, the meta-analysis will be presented using pooled estimates with 95% confidence interval as well as the estimate of the between study variance. The studies reporting similar measures will be grouped accordingly, i.e., hazard ratios (HRs), odds ratios (ORs) and risk ratios (RRs). Those reporting adjusted and crude measures will also be grouped separately. Where data is available, mean difference (MD) with its standard deviation (SD) for continuous data such as VL will be calculated while ORs or RRs will be used for categorical data. Statistical heterogeneity assessment will be done using the I^2 statistic [35], by examining the results using the chi-squared test with p≤0.10. We will consider I^2 statistical values between 30% to 60% as moderate heterogeneity; >50% to 90% as substantial heterogeneity; and ≥75% as considerable heterogeneity. It is important to note that the I^2statistic depends on the direction and magnitude of effects across several studies as well as the strength of evidence for heterogeneity [35]. A random effects model will be used. We will conduct subgroup analysis by study designs and study settings (i.e., country) where substantial heterogeneity exists. To determine publication bias, a Funnel Plot and Egger's test will be used. All analyses will be performed using STATA 16 [36].

In a case where quantitative synthesis is not possible, information will be presented in text format as well as tables to summarise and present the characteristics and findings of the studies in a narrative systematic synthesis to explore the relationship between lack of knowledge of male partner's HIV status and maternal and child health outcomes as per data from the included studies. This synthesis will be guided by the Centre for Reviews and Dissemination [37].

Quality of evidence

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) working group methodology will be used to judge the quality of evidence for all outcomes. The following domains will be used for this assessment, "risk of bias, inconsistency, indirectness, imprecision and publication bias". Given this is a prognostic type research question unlike in therapeutic studies, observational studies will be ranked with higher certainty of evidence compared to RCTs. The higher confidence in observational studies for prognostic factor studies is due to the stricter eligibility criteria in RCTs which is likely to exclude the population of interest for the prognostic question [38]. The significance levels of evidence for the risk

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

associated with not knowing the HIV status of a male partner will be ranked as high, moderate, low, or very low [39].

Presenting and reporting of results

We plan to display the study selection process by means of a flow diagram and give reasons for exclusion of studies. Reporting of the results will follow PRISMA guidelines [30]. We will display the search strings, risk of bias tables, forest plots and summary of study findings tables.

Patient and Public involvement

No patient involved.

Ethics and dissemination

The study will use publicly available data and ethics exemption has been obtained from Human Research Ethics Committees, Faculty of Medicine & Health Sciences, Stellenbosch University Findings of this systematic review will be disseminated in peer-review journals including various media platforms i.e. webinars, symposia, conferences or congresses

Conclusion

This systematic review will provide an overview of how lack of disclosure of male partner's HIV status to pregnant and postpartum women affects progress towards eliminating HIV incidence and reducing HIV morbidity and mortality among pregnant women and postpartum women and their infants. The study will also help to highlight knowledge gaps in the WHO priority countries about the effect of knowing the HIV status of a male partner on improving outcomes of PMTCT program

Declarations

Author contributions: This systematic review was conceived and drafted by NKN and TMM. Subsequent drafts were reviewed by NKN, OA, PN. All authors approved final version.

Funding Statement: This systematic review received no funding from either the public, commercial or non-profit sectors. NKN and TMM are supported by the South African Medical Research Council.

Competing interests: None declared.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

References

1. The Joint United Nations Programme on HIV/AIDS (UNAIDS). Factsheet World AIDS Day. Global HIV Statistics. 2020. https://www.unaids.org/en/resources/fact-sheet date accessed 26 January 2021
2. The United Nations International Children's Emergency Fund. Global and Regional trends. July 2020. 2020. https://data.unicef.org/topic/hivaids/global-regional-trends/ date accessed 28 May 2021
3. UNAIDS. A Global Plan Towards the Elimination of New HIV Infections among Children by 2015 and keeping their Mothers Alive (2011-2015). 2011. https://files.unaids.org/en/media/unaids/contentassets/documents/unaidspublication/2011/20110609_JC2137_Global-Plan-Elimination-HIV-Children_en.pdf date accessed 30 May 2021
4. World Health Organisation. Use of Antiretroviral Drugs for treating Pregnant Women and preventing HIV infections in Infants. 2012. https://www.who.int/hiv/PMTCT_update.pdf date accessed 26 January 2020
5. Tam, M., Amzel, A, & Phelps, BR. Disclosure of HIV serostatus among pregnant and postpartum women in sub-Saharan Africa: a systematic review. *AIDS Care* 2015;4: 436-450.
6. Barr, BAT, van Lettow, M, van Oosterhout, JJ, et al. National estimates and risk factors associated with early mother-to-child transmission of HIV after implementation of option B+: a cross-sectional analysis. *Lancet HIV* 2018;5:688-695.
7. Kuonza LR, Tshuma CD, Shambira GN, et al. Non-adherence to the single dose nevirapine regimen for the prevention of mother-to-child transmission of HIV in Bindura town, Zimbabwe: a cross-sectional analytic study. *BMC Public Health*2010;10(1):1-8.
8. Cohen MS, Chen YQ, McCauley M, et al. Prevention of HIV-1 infection with early antiretroviral therapy. *N Engl J Med*2011;365(6):493-505.
9. Davis NL, Miller WC, Hudgens MG, et al. Maternal and Breast Milk Viral Load: Impacts of Adherence on Peri-Partum HIV Infections Averted-the BAN Study. *J Acquir Immune Defic Syndr*2016;73(5):572.
10. Mnyani CN, Simango A, Murphy J,et al. Patient factors to target for elimination of mother-to-child transmission of HIV. *Global Health*2014;10(1):1-7.
11. Brittain K, Phillips TK, Zerbe A, et al. Long-term effects of unintended pregnancy on antiretroviral therapy outcomes among South African women living with HIV. *AIDS*2019;33(5):885.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Chetty T, Vandormael A, Thorne C, et al. Incident HIV during pregnancy and early postpartum period: a population-based cohort study in a rural area in KwaZulu-Natal, South Africa. *BMC Pregnancy and Childbirth*2017;17(1):1-0.
13. De Schacht C, Mabunda N, Ferreira Jr OC, et al. High HIV incidence in the postpartum period sustains vertical transmission in settings with generalized epidemics: a cohort study in Southern Mozambique. *J Int AIDS Soc* 2014;17(1):18808.
14. van Lettow M, Cataldo F, Landes M, et al. Impact of inter-partner HIV disclosure patterns in Malawi's PMTCT program: A mixed-method study. *PloS One* 2019;14(7):e0219967.
15. Colombini M, Stöckl H, Watts C, et al. Factors affecting adherence to short-course ARV prophylaxis for preventing mother-to-child transmission of HIV in sub-Saharan Africa: a review and lessons for future elimination. *AIDS Care*2014 Jul;26(7):914-26.
16. Chirinda W, Ramraj T, Singh Y, et al. Knowledge of partner HIV serostatus and disclosure among South African women with recent pregnancy, 2011-2012. *J Glob Health* Accepted May 2021.
17. Kalua T, Barr BA, Van Oosterhout JJ, et al. Lessons learned from option B+ in the evolution toward "test and start" from Malawi, Cameroon, and the United Republic of Tanzania. *J Acquir Immune Defic Syndr*2017;75(Suppl 1):S43.
18. Yitayew YA, Bekele DM, Demissie BW, et al. Mother to child transmission of HIV and associated factors among HIV exposed infants at public health facilities, Dessie Town, Ethiopia. *HIV/AIDS*2019;11:343.
19. Kiweewa FM, Bakaki PM, McConnell MS, et al. A cross-sectional study of the magnitude, barriers, and outcomes of HIV status disclosure among women participating in a perinatal HIV transmission study,"the Nevirapine Repeat Pregnancy study". *BMC Public Health*2015;15(1):1-7.
20. Nordberg B, Gabriel EE, Were E, et al. Social concerns related to HIV status disclosure and participation in the prevention of mother-to-child transmission of HIV care among pregnant women in Kenya. *BMC Pregnancy and Childbirt*h2020;20:1-9.
21. Odiachi A, Erekaha S, Cornelius LJ, et al. HIV status disclosure to male partners among rural Nigerian women along the prevention of mother-to-child transmission of HIV cascade: a mixed methods study. *Reprod Health*2018;15(1):1-2.
22. Bucagu M, de Dieu Bizimana J, Muganda J, et al. Socio-economic, clinical and biological risk factors for mother-to–child transmission of HIV-1 in Muhima health centre (Rwanda): a prospective cohort study. *Arch Public Health*2013;71(1):1-2.
23. Brittain K, Mellins CA, Remien RH, et al. Impact of HIV-status disclosure on HIV viral load in pregnant and postpartum women on antiretroviral therapy. *J Acquir Immune Defic Syndr*2019;81(4):379-86.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

24. Manjate Cuco RM, Munguambe K, Bique Osman N, et al. Male partners' involvement in prevention of mother-to-child HIV transmission in sub-Saharan Africa: a systematic review. *SAHARA-J*2015;12(1):87-105.
25. Chanyalew H, Girma E, Birhane T, et al. Male partner involvement in HIV testing and counseling among partners of pregnant women in the Delanta District, Ethiopia. *Plos One*2021;16(3):e0248436.
26. Osoti AO, John-Stewart G, Kiarie J, et al. Home visits during pregnancy enhance male partner HIV counseling and testing in Kenya: a randomized clinical trial. *AIDS (London, England)*2014;28(1):95.
27. Dinh TH, Delaney KP, Goga A, et al. Impact of maternal HIV seroconversion during pregnancy on early mother to child transmission of HIV (MTCT) measured at 4-8 weeks postpartum in South Africa 2011-2012: a national population-based evaluation. *PloS One*2015;10(5):e0125525.
28. UNAIDS. Condom use declining. 2020. Date accessed 9 June 2021 https://www.unaids.org/en/keywords/condoms
29. Shikwane ME, Villar-Loubet OM, Weiss SM, et al. HIV knowledge, disclosure and sexual risk among pregnant women and their partners in rural South Africa. *SAHARA-J*2013;10(2):105-12.
30. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev*2015;4(1):1-9.
31. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan-a web and mobile app for systematic reviews. Syst Rev. 2016. org
32. Higgins JP, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ*2011;343:1–9. https:// doi.org/10.1210/jcem-43-6-1203.
33. Sterne JAC, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomized studies of interventions. *BMJ*2016;355;i4919; doi: 10.1136/bmj.i4919.
34. Hayden JA, Coté P, Bombardier C. Evaluation of the Quality of Prognosis Studies in Systematic Reviews. *Ann Intern Med*2006;144(6):427-438
35. Higgins JP, Thompson SG, Deeks JJet al. Measuring inconsistency in meta-analyses. *BMJ*2003;327(7414):557.
36. StataCorp. 2019. Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC.
37. Akers J, Aguiar-Ibáñez R, Baba-Akbari A. Systematic reviews: CRD's guidance for undertaking reviews in health care.

For peer review only

38. Iorio A, Spencer FA, Falavigna M, et al. Use of GRADE for assessment of evidence about prognosis: rating confidence in estimates of event rates in broad categories of patients. *BMJ*2015;350.
39. Foroutan F, Guyatt G, Zuk V, et al. GRADE Guidelines 28: Use of GRADE for the assessment of evidence about prognostic factors: rating certainty in identification of groups of patients with different absolute risks. *J Clinl Epidemiol*2020;121:62-70.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Figures

Figure 1: A summary of reported and potential role of male-partner HIV disclosure within the PMTCT causal pathways.

Figure 2: Flowchart showing detailed summary of the search and screening processes

Supplementary material

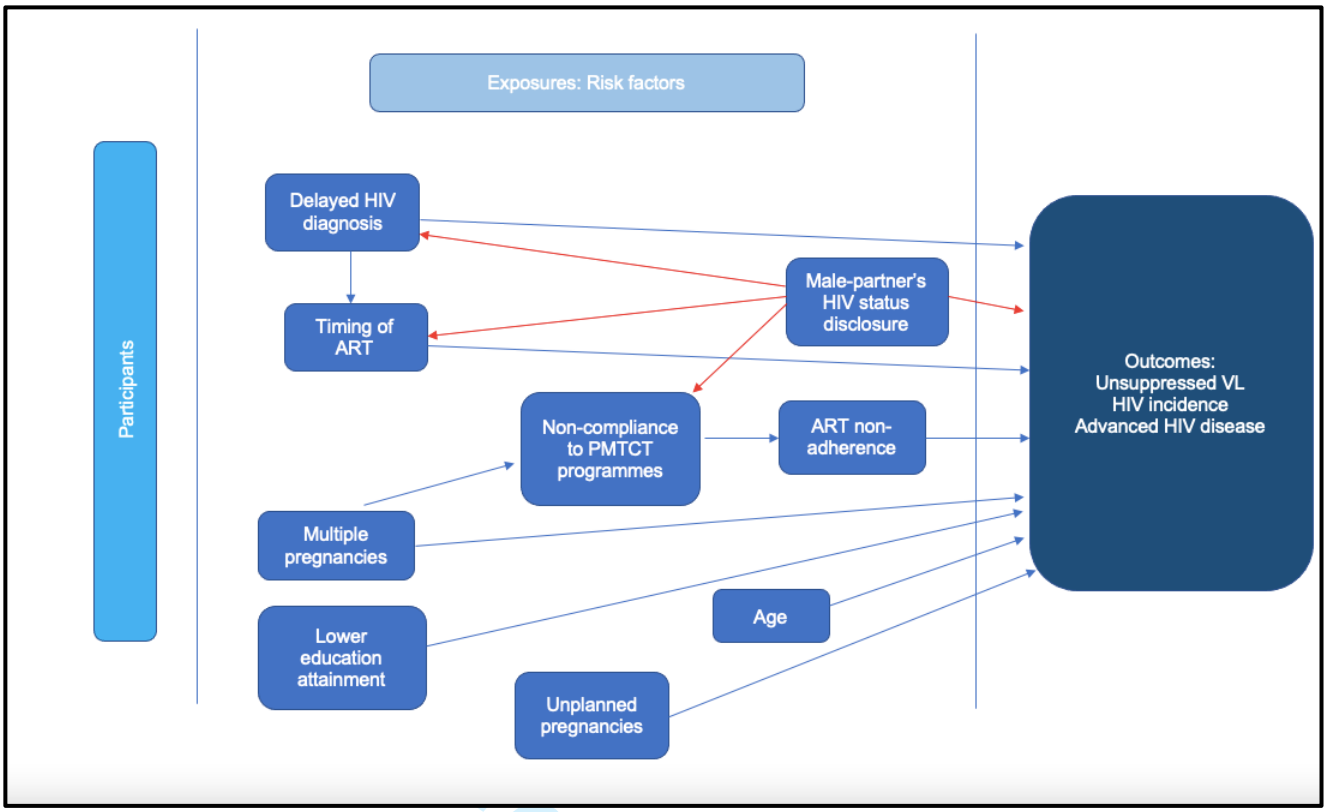
Supplementary file 1:

Title: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol

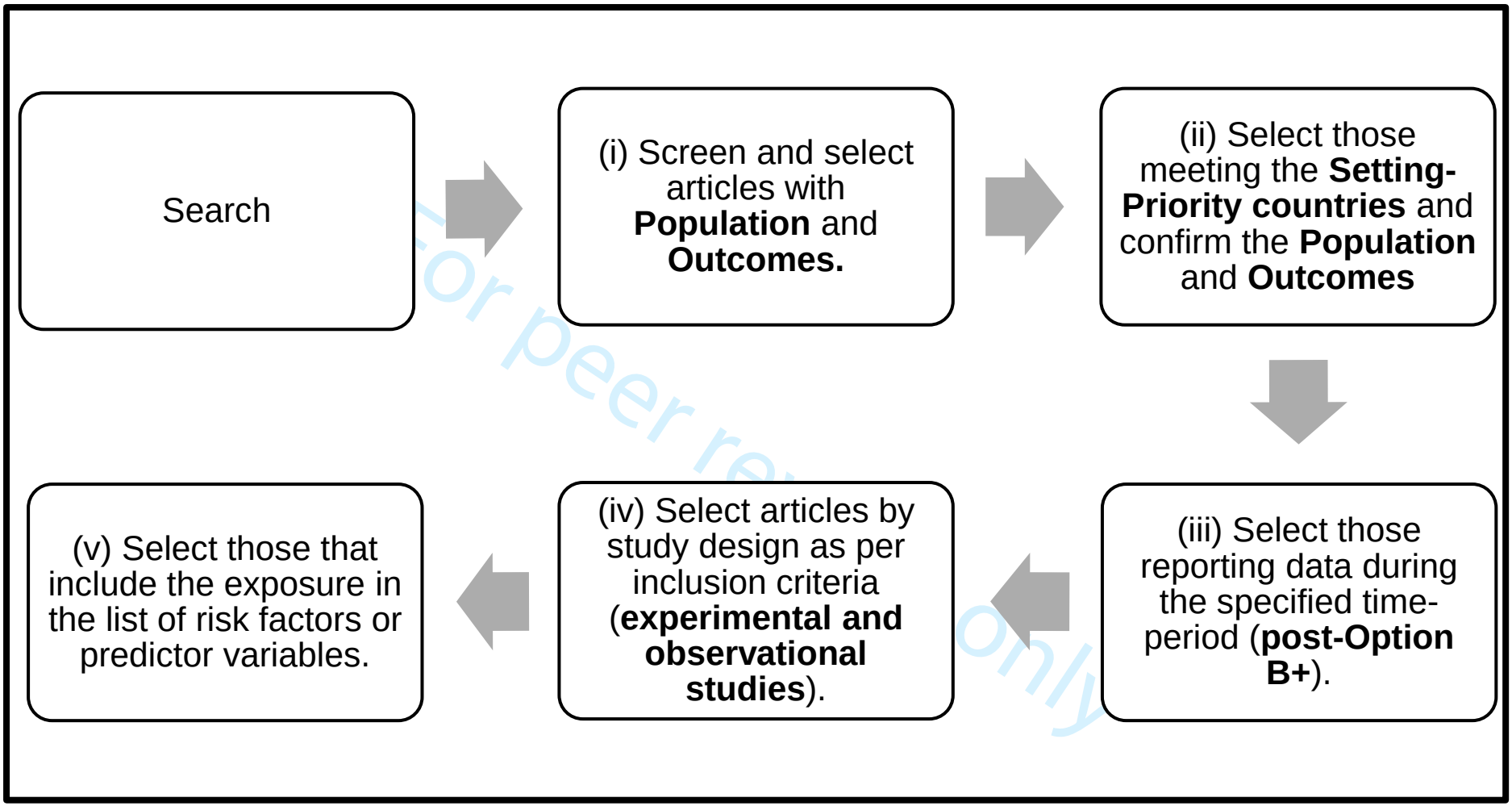
Format: PDF

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml



For peer review only



For peer review only

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page no.
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	6
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	6
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	10
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3-5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records:			
Data	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-8

For peer review only

management			
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

For peer review only

BMJ Open

Male partner unknown HIV status as a risk factor for HIV incidence and clinical outcomes in Prevention of Mother-to-Child Transmission of HIV programmes in 21 WHO priority countries, a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-057190.R1
Article Type:	Protocol
Date Submitted by the Author:	21-Jan-2022
Complete List of Authors:	Mmotsa, Tshiamo; South African Medical Research Council, HIV Prevention Research Unit Ngandu, Nobubelo; South African Medical Research Council, HIV Prevention Research Unit Adetokunboh, Olatunji; Stellenbosch University, DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA) Nyasulu, P; Stellenbosch University, Division of Epidemiology and Biostatistics
Primary Subject Heading:	Epidemiology
Secondary Subject Heading:	HIV/AIDS
Keywords:	EPIDEMIOLOGY, HIV & AIDS < INFECTIOUS DISEASES, Community child health < PAEDIATRICS

SCHOLARONE™
Manuscripts

Male partner unknown HIV status as a risk factor for HIV incidence and clinical outcomes in Prevention of Mother-to-Child Transmission of HIV programmes in 21 WHO priority countries, a systematic review protocol

Tshiamo Moshading Mmotsa: HIV Prevention Research Unit, South African Medical Research Council, Cape Town, South Africa

Nobubelo Kwanele Ngandu (corresponding author): HIV Prevention Research Unit, South African Medical Research Council, P. O. Box 19070, Tygerberg 7505, Cape Town, South Africa. Email: Nobubelo.Ngandu@mrc.ac.za

Olatunji O Adetokunboh: DSI-NRF Centre of Excellence in Epidemiological Modelling and Analysis (SACEMA), Stellenbosch University, Stellenbosch, South Africa. Division of Epidemiology and Biostatistics, Department of Global Health, Stellenbosch University, Cape Town, South Africa. Health Sciences Department, The University of the People, Pasadena, United States.

Peter Nyasulu: Division of Epidemiology and Biostatistics, Faculty of Medicine and Health Sciences, Stellenbosch University, Cape Town, South Africa. Division of Epidemiology and Biostatistics, School of Public Health, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa

Word count = 2607

For peer review only

Abstract

Introduction Research has shown an association between increased disclosure of Human Immunodeficiency Virus (HIV) status by pregnant and breastfeeding women and improved clinical health and that of their infant. Increasing awareness about their male partner's HIV status will no doubt lead to even better outcomes at the population level. Male partner involvement is important for improving outcomes of prevention of mother-to-child transmission of HIV (MTCT) as it improves social support and commitment from both parents of the baby to ensure sustained good health. Although lack of knowledge of the HIV status of a male partner is of great concern, limited research has been done to determine whether it remains one of the barriers to reaching the proposed goals of eliminating MTCT in pregnant or postpartum women. Our aim is to determine if lack of knowledge of a male partner's HIV status is a significant risk factor for HIV incidence and poor HIV clinical outcomes among pregnant women and postpartum women and their infants.

Methods and analysis A systematic review and meta-analysis of experimental and observational studies will be conducted. The review will focus on knowledge of male partner's HIV status in the 21 priority countries most affected by HIV in Africa. We will search electronic databases such as PubMed/MEDLINE, Scopus, Web of Science and Cochrane library, Science Direct, CINAHL, LILACS and SciELO databases from January 2011 to December 2021. We will also search the Pan African and WHO clinical trial registries and conference archives. We will conduct a quality assessment of eligible studies and evaluate the heterogeneity of the pooled studies using the I^2 statistic. The statistical analysis will be performed using STATA statistical software version 16.

Ethics and dissemination The study will use publicly available data and ethics exemption has been obtained from Human Research Ethics Committees, Faculty of Medicine & Health Sciences, Stellenbosch University. The protocol was registered on Prospective Register of Systematic Reviews, registration number CRD42021247686, in May 2021. Findings of this systematic review will be disseminated in peer-review journals including various media platforms i.e., webinars, symposia, conferences, or congresses.

Keywords male-partner, HIV disclosure, mother-to-child transmission of HIV, pregnant and postpartum, breastfeeding women, systematic review

For peer review only

Prospective Register of Systematic Reviews, registration number CRD42021247686.

This is an open access article distributed in accordance with the Creative Commons Attribution Non-Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: http://creativecommons.org/licenses/by-nc/4.0/.

Strengths and limitations of this study

- This study protocol outlines a path that will minimise data duplication and provides clear and transparent methods used.
- This study will provide quality evidence through a rigorous methodological process which follows the PRISMA-P guidelines including meta-analysis and quality assessment of studies using the ROBINS-I and QUIPS tools.
- This systematic review will arguably be the first to review the role of Male partner unknown HIV status on HIV incidence and clinical outcomes in PMTCT programmes in 21 WHO priority countries.
- The findings are expected to highlight urgent research gaps around the role and impact of HIV non-disclosure amongst male partners on progress towards achieving HIV elimination targets in PMTCT programmes.
- There is potential limitation arising from publication bias due to inclusion of English only studies and methodological inability to include qualitative studies.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Introduction

Burden of HIV in mothers and children

For almost two decades, it has been a top priority to eliminate HIV incidence amongst pregnant and postpartum women and their children. [1, 2, 3]. In 2010, the World Health Organization (WHO) recommended provision of lifelong antiretroviral therapy (ART) to pregnant women regardless of CD4 count, known as the Prevention of Mother-to-Child Transmission of HIV (PMTCT) Option B+ policy [4]. Concurrently, 21 countries which historically had limited access to CD4 testing services and partner testing, high fertility rates, prolonged breastfeeding periods and a high generalized HIV epidemic, were prioritized for HIV elimination targets. These countries also accounted for 90% of the world's HIV positive cases in pregnant women and in a 2020 report most of these countries accounted for more than two thirds up to one fifth of the newly infected children [4, 5]. Despite lifelong ART under the Option B+ regimen, the PMTCT programmes have been slower than anticipated in eliminating HIV incidence in mothers and children. A 2020 report has shown that even with increased ART coverage among pregnant and breastfeeding women in the 21 priority countries, mother-to-child transmission of HIV (MTCT) ranged from 2% (in Botswana) to 28% in the Democratic Republic of Congo [5]. Several factors are responsible for the delay in meeting the HIV elimination targets. Uptake of ART and timing of initiation are very important in the PMTCT cascade as they affect the effectiveness of the programme [6, 7] and delayed initiation is likely to increase maternal viral load thus increasing risk for MTCT [8]. Additionally, unplanned pregnancies, multiple pregnancies and low education attainment have also been reported as risk factors for MTCT as well as poor ART outcomes [9-11]. Younger and adolescent women are also more likely to vertically transmit HIV to their infants because of their increased likelihood of poor ART outcomes [12, 13]. Fear of disclosure is another risk factor as it may lead to intimate partner violence and reduced partner support for PMTCT or ART initiation [14].

Disclosure of HIV status is a complex risk factor in PMTCT programmes

Disclosure of HIV status is a complex risk factor for not achieving PMTCT targets, given co-presence of other related risk factors such as women not knowing their own HIV status or their male partner's HIV status as well as low partner testing,[9, 13, 15, 16]. Since the end of 2014, pregnant women HIV disclosure rates have been increasing. Results from a meta-analysis found that more than 67% of pregnant women in sub-Saharan Africa had disclosed their HIV status [17]. Increased disclosure rates were also evident in subsequent studies conducted after 2015, with disclosure rates of 80%, 85%, 94.9% and 99% in Kenya, Nigeria, Malawi and Uganda, respectively [6, 18-21] and disclosure to male partners was more evident than

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

disclosure to family members [6, 17, 19]. According to Bucagu et al., [21] HIV status disclosure is more frequent in women than in males and disclosure is generally higher in steady partners than in casual partners. HIV positive status non-disclosure of both male partners and pregnant or postpartum women was a strong predictor of suboptimal ART adherence and MTCT [13]. A systematic review on factors that influence maternal and infant ART adherence found that HIV status non-disclosure in HIV positive pregnant and postpartum women to their male partners or family was one of the factors resulting in sub-optimal ART adherence [11]. Non-adherence to ART and poor ART outcomes were significantly associated with increased proportions of non-disclosure among pregnant and postpartum women [6, 21, 22]. Non-disclosure to either male partners, family or co-workers made it difficult for women to comply with attendance or participation in PMTCT programmes resulting in greater rates of loss-to-follow-up and poorer ART outcomes [21, 23]. This shows that disclosure, especially to male partners, is important for improved ART outcomes thus prevention of HIV mortality and reducing HIV incidence in infants as well as keeping their mothers healthy but it remains unclear whether knowing their male partner's HIV status has also improved over the years.

Research on knowing the male partner's HIV status

The uptake of HIV Testing Services (HTS) among men is generally low, and research has shown that male partners of pregnant or postpartum women have no concerns when it comes to knowing their HIV status or with HIV prevention [24]. A 2018 study conducted in the Delanta District in Ethiopia found that men also had fears when it comes to testing for HIV, only 57% males were involved in HTS [25]. In a facility-based study conducted between August 2011 and March 2012 across all nine provinces in South Africa, the prevalence of not knowing a male partner's HIV status amongst pregnant and breastfeeding women, was reported to be as high as 34%[26]. Given evidence of an association between increased disclosure of pregnant and breastfeeding women's HIV status and improvement in their HIV clinical health and that of their infants [21], increasing awareness about their male partner's HIV status will no doubt lead to even better outcomes at the population level. Figure 1 shows a summary of reported and potential (red arrows) role of male-partner HIV disclosure within the PMTCT causal pathways. Although not knowing a male partner's HIV status is a potential concern, limited research has been done to determine whether it remains one of the barriers to reaching the proposed goals to eliminate MTCT and HIV in pregnant or postpartum women. This systematic review and meta-analysis aim to determine whether not knowing a male partner's HIV status is a significant risk factor for HIV incidence and poor HIV clinical outcomes among pregnant and postpartum women and their infants in the 21 WHO priority countries for eliminating MTCT.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Methods and analysis

The protocol has been developed in accordance with the 2015 Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA-P) (Supplementary file 1) [27]. This systematic review is registered with the United Kingdom's National Institute of Health Research (NIHR) Prospective Register of Systematic Reviews (PROSPERO) registration number CRD42021247686.

Inclusion criteria

All studies that report the knowledge of the HIV status of the male partner as an exposure or predictor factor for HIV incidence, viral load or CD4 count or WHO HIV clinical disease stage during pregnancy or postpartum period among women or their infants aged 0-24 months from the 21 priority countries most affected by HIV in Africa (Table 1), will be reviewed. The studies should also have been conducted after the adoption of the PMTCT Option B+ regimen in these listed countries (from the year 2011). We will include English language or articles translated into the English language. Studies reporting measures of association, i.e., experimental (randomized control trials and non-randomized trials) and observational studies (case-control, cohort, and cross-sectional studies) will all be included.

Exclusion criteria

Publications not from the priority countries listed in Table 1 or reporting outcomes from the option A or option B policies, or outcomes that are not HIV incidence, viral load, CD4 count, and WHO HIV clinical disease staging will not be included. Qualitative studies will not be included.

Table 1: WHO/UNAIDS global priority countries and adoption of PMTCT Option B+ regimen [4].

Countries Included	
Angola	Mozambique
Botswana	Namibia
Burundi	Nigeria
Cameroon	Tanzania
Chad	The Democratic Republic of the Congo
Côte d'Ivoire	South Africa

For Peer Review Only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Ethiopia	Swaziland
Ghana	Uganda
Kenya	Zambia
Lesotho	Zimbabwe
Malawi	

Information sources and search strategy

A comprehensive search will be conducted in the following databases: PubMed (via MEDLINE), the Cochrane Library, Scopus, Web of Science, Science Direct, CINAHL, LILACS and SciELO. Additional searches will be conducted from the Pan African Clinical Trials Registry (PACTR) (https://pactr.samrc.ac.za/), International Clinical Trials Registry Platform (ICTRP) (https://www.who.int/clinical-trials-registry-platform), clinicaltrials.gov as well as published conference abstracts from the International AIDS Society (IAS) Conference, Conference on Retroviruses and Opportunistic Infections (CROI) and the International Conference on AIDS and STIs in Africa. Reference lists of included studies will also be reviewed to identify any eligible studies that could be added.

The systematic search will be done using various combinations of synonyms of the key search terms that define the Population and the overarching Outcome from the Participant/Exposure/Comparison/Outcome/Time-period/Setting (PECOTS) elements [28] of the research question (Table 2). Search terms will include: 'HIV' or 'AIDS' or 'MTCT' for the Outcome and 'Pregnant' or 'Breastfeeding' or 'Postpartum women' for the Population. A pilot exercise will be conducted to identify other synonyms and search terms/phrases needed to develop an exhaustive search strategy (see preliminary search strategy in Supplementary file 2). All studies found will be uploaded into RAYYAN—a systematic review managing software for further processing [29] and duplicates articles will be removed.

Table 2: The PECOTS framework

P- Participants	HIV positive pregnant and breastfeeding women, HIV negative pregnant and breastfeeding women, Infants born to HIV positive women
E- Exposure	Lack of knowledge of male partner's HIV status
C- Comparison	Knowledge of male partner's HIV status
O- Outcome	Unsuppressed viral load, HIV incidence, Advanced HIV disease (defined using CD4 count or WHO clinical staging)
T- Time period	Post- adoption of the PMTCT Option B+ regimen in that country
S- Setting	21 WHO priority countries

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Selection process

The selection of studies will be done in duplicate by two reviewers (TM, NN) and discrepancies will be resolved by discussing with a third reviewer (PN, OA). The screening and selection of the manuscripts will be done in several steps and informed by the inclusion criteria of the PECOTS [27].

Proposed steps to guide the screening and selection process of manuscripts that will be uploaded on to the RAYYAN system are presented in Figure 2 in the following order: (i) titles and abstracts will be screened for the research question's Population and Outcomes, full text articles from this step will be downloaded; articles with the following characteristics will then be selected: (ii) meets the study Setting (WHO priority country) and the Population and Outcomes of the inclusion criteria; (iii) publishing data from the defined Time period (during Option B+ policy); (iv) meet the study design as per inclusion criteria (experimental and observational studies); and (v) have included the Exposure of interest in the data as a predictor/exposure variable.

Data Extraction

Data from the selected full-text manuscripts will be extracted independently by the two reviewers (TM, NN) using a standardised piloted extraction form. Any discrepancies will be resolved by an agreement between the two reviewers or the third if there is no resolution (OA, PN). In cases where additional information is required such as clarifying on information, missing information or regarding relevant data that may not have been included in the manuscript, the authors will contact the corresponding author of the primary studies to access the information [27].

The following data parameters will be extracted; author, title of the study, study setting, study design, study sample and population, year of study, measure of association or effect between the exposure of interest and HIV incidence/clinical disease.

Quality of manuscripts

Study quality and risk of bias will be assessed by evaluating the study design, data collection as well as the data analysis methods. The quality assessment will be double-checked by the review team. The risk of bias of randomised studies will be assessed using the Cochrane Risk of Bias Tool [30]. The Risk of Bias for Non-Randomised Studies- of Interventions (ROBINS-I) Tool [31] will be used for non-randomised trials while the Quality in Prognostic Studies (QUIPS) Tool [32] will be used for observational studies. The summary of characteristics of each study will be presented in a table format.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Synthesis of results

A meta-analysis will be performed if the selected manuscripts are homogenous in terms of methodology and outcomes. Data that is sufficiently similar will be pooled together using the inverse variance approach to accommodate crude and adjusted measures of association and effect, where possible [27]. Additionally, the meta-analysis will be presented using pooled estimates with 95% confidence interval as well as the estimate of the between study variance [27]. The studies reporting similar measures will be grouped accordingly, i.e., hazard ratios (HRs), odds ratios (ORs) and risk ratios (RRs). Those reporting adjusted and crude measures will also be grouped separately. Where data is available, mean difference (MD) with its standard deviation (SD) for continuous data such as VL will be calculated while ORs or RRs will be used for categorical data. Statistical heterogeneity assessment will be done using the I^2 statistic [33], by examining the results using the chi-squared test with p≤0.10. We will consider I^2 statistical values between 30% to 60% as moderate heterogeneity; >60% to 90% as substantial heterogeneity; and >90 to 100% as considerable heterogeneity. It is important to note that the I^2statistic depends on the direction and magnitude of effects across several studies as well as the strength of evidence for heterogeneity [33]. A random effects model will be used. We will conduct subgroup analysis by study designs and study settings (i.e., country) where substantial heterogeneity exists. To determine publication bias, a Funnel Plot and Egger's test will be used. All analyses will be performed using STATA 16 [34].

In a case where quantitative synthesis is not possible, information will be presented in text format as well as tables to summarise and present the characteristics and findings of the studies in a narrative systematic synthesis to explore the relationship between lack of knowledge of male partner's HIV status and maternal and child health outcomes as per data from the included studies. This synthesis will be guided by the Centre for Reviews and Dissemination [35].

Quality of evidence

The Grading of Recommendations Assessment, Development and Evaluation (GRADE) working group methodology will be used to judge the quality of evidence for all outcomes [36]. The following domains will be used for this assessment, "risk of bias, inconsistency, indirectness, imprecision and publication bias". Given this is a prognostic type research question unlike in therapeutic studies, observational studies will be ranked with higher certainty of evidence compared to RCTs. The higher confidence in observational studies for prognostic factor studies is due to the stricter eligibility criteria in RCTs which is likely to exclude the population of interest for the prognostic question [36]. The significance levels of

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

evidence for the risk associated with not knowing the HIV status of a male partner will be ranked as high, moderate, low, or very low [37].

Presenting and reporting of results

We plan to display the study selection process by means of a flow diagram and give reasons for exclusion of studies. Reporting of the results will follow PRISMA guidelines [27]. We will display the search strings, risk of bias tables, forest plots and summary of study findings tables [27].

Patient and Public involvement

No patient involved.

Ethics and dissemination

The study will use publicly available data and ethics exemption has been obtained from Human Research Ethics Committee, Faculty of Medicine & Health Sciences, Stellenbosch University. Findings of this systematic review will be disseminated in peer-review journals including various media platforms i.e. webinars, symposia, conferences, or congresses.

Discussion

This systematic review will provide an overview of how lack of disclosure of male partner's HIV status to pregnant and postpartum women affects progress towards eliminating HIV incidence and reducing HIV morbidity in pregnant and postpartum women and their infants. This systematic review will arguably be the first to explore the risk of male partner unknown HIV status on HIV incidence and clinical outcomes in PMTCT programmes in the 21 WHO priority countries. As a result of a clearly outlined protocol, there will be minimum duplication of data. This protocol also ensures that a clear and transparent procedure is used. Some limitations are foreseen in this study, such as the non-inclusion of qualitative studies in the meta-analysis methodology to be used, potentially low methodological quality of studies from grey literature (e.g., policy reports) and low statistical power for meta-analysis where few included studies report on the same outcome. However, the study will still help to highlight knowledge gaps in the WHO priority countries about the role of knowing a male partner's HIV status in improving PMTCT program effectiveness. There is generally scarcity in randomized and non-randomized trials assessing the impact of male involvement in MTCT of HIV programmes as well as studies specifically focused on the impact of lack of knowledge of male partner HIV status on maternal as well as child outcomes. This systematic review will therefore

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

likely pave the way for more primary studies to explore male partner HIV status disclosure as a possible risk factor.

Declarations

Author contributions: This systematic review was conceived and drafted by NKN and TMM. Subsequent drafts were reviewed by NKN, OA, PN. All authors approved final version.

Funding Statement: This systematic review received no funding from either the public, commercial or non-profit sectors. NKN and TMM are supported by the South African Medical Research Council.

Competing interests: None declared.

For peer review only

References

1. The Joint United Nations Programme on HIV/AIDS (UNAIDS). Factsheet World AIDS Day. Global HIV Statistics. 2020. https://www.unaids.org/en/resources/fact-sheet date accessed 26 January 2021
2. The United Nations International Children's Emergency Fund. Global and Regional trends. July 2020. 2020. https://data.unicef.org/topic/hivaids/global-regional-trends/ date accessed 28 May 2021
3. UNAIDS. A Global Plan Towards the Elimination of New HIV Infections among Children by 2015 and keeping their Mothers Alive (2011-2015). 2011. https://files.unaids.org/en/media/unaids/contentassets/documents/unaidspublication/2011/20110609_JC2137_Global-Plan-Elimination-HIV-Children_en.pdf date accessed 30 May 2021
4. World Health Organisation. Use of Antiretroviral Drugs for treating Pregnant Women and preventing HIV infections in Infants. 2012. https://www.who.int/hiv/PMTCT_update.pdf date accessed 26 January 2020
5. UNAIDS. Start Free, Stay Free, AIDS Free. Final report on 2020 targets: July 2021. https://www.aidsdatahub.org/resource/start-free-stay-free-aids-free-final-report-2020-targets date accessed 06 January 2021
6. Barr, BAT, van Lettow, M, van Oosterhout, JJ, et al. National estimates and risk factors associated with early mother-to-child transmission of HIV after implementation of option B+: a cross-sectional analysis. *Lancet HIV* 2018;5:688-695.
7. Cohen MS, Chen YQ, McCauley M, et al. Prevention of HIV-1 infection with early antiretroviral therapy. *N Engl J Med* 2011;365(6):493-505.
8. Davis NL, Miller WC, Hudgens MG, et al. Maternal and Breast Milk Viral Load: Impacts of Adherence on Peri-Partum HIV Infections Averted-the BAN Study. *J Acquir Immune Defic Syndr* 2016;73(5):572.
9. Mnyani CN, Simango A, Murphy J,et al. Patient factors to target for elimination of mother-to-child transmission of HIV. *Global Health* 2014;10(1):1-7.
10. Brittain K, Phillips TK, Zerbe A, et al. Long-term effects of unintended pregnancy on antiretroviral therapy outcomes among South African women living with HIV. *AIDS* 2019;33(5):885.
11. Colombini M, Stöckl H, Watts C, et al. Factors affecting adherence to short-course ARV prophylaxis for preventing mother-to-child transmission of HIV in sub-Saharan Africa: a review and lessons for future elimination. *AIDS Care* 2014 Jul;26(7):914-26.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

12. Chetty T, Vandormael A, Thorne C, et al. Incident HIV during pregnancy and early postpartum period: a population-based cohort study in a rural area in KwaZulu-Natal, South Africa. *BMC Pregnancy and Childbirth* 2017;17(1):1-0.
13. De Schacht C, Mabunda N, Ferreira Jr OC, et al. High HIV incidence in the postpartum period sustains vertical transmission in settings with generalized epidemics: a cohort study in Southern Mozambique. *J Int AIDS Soc* 2014;17(1):18808.
14. van Lettow M, Cataldo F, Landes M, et al. Impact of inter-partner HIV disclosure patterns in Malawi's PMTCT program: A mixed-method study. *PloS One* 2019;14(7):e0219967.
15. Kalua T, Barr BA, Van Oosterhout JJ, et al. Lessons learned from option B+ in the evolution toward "test and start" from Malawi, Cameroon, and the United Republic of Tanzania. *J Acquir Immune Defic Syndr* 2017;75(Suppl 1):S43.
16. Yitayew YA, Bekele DM, Demissie BW, et al. Mother to child transmission of HIV and associated factors among HIV exposed infants at public health facilities, Dessie Town, Ethiopia. *HIV/AIDS* 2019;11:343.
17. Tam, M., Amzel, A, & Phelps, BR. Disclosure of HIV serostatus among pregnant and postpartum women in sub-Saharan Africa: a systematic review. *AIDS Care* 2015;4: 436-450.
18. Kiweewa FM, Bakaki PM, McConnell MS, et al. A cross-sectional study of the magnitude, barriers, and outcomes of HIV status disclosure among women participating in a perinatal HIV transmission study,"the Nevirapine Repeat Pregnancy study". *BMC Public Health* 2015;15(1):1-7.
19. Nordberg B, Gabriel EE, Were E, et al. Social concerns related to HIV status disclosure and participation in the prevention of mother-to-child transmission of HIV care among pregnant women in Kenya. *BMC Pregnancy and Childbirth* 2020;20:1-9.
20. Odiachi A, Erekaha S, Cornelius LJ, et al. HIV status disclosure to male partners among rural Nigerian women along the prevention of mother-to-child transmission of HIV cascade: a mixed methods study. *Reprod Health* 2018;15(1):1-2.
21. Bucagu M, de Dieu Bizimana J, Muganda J, et al. Socio-economic, clinical and biological risk factors for mother-to–child transmission of HIV-1 in Muhima health centre (Rwanda): a prospective cohort study. *Arch Public Health* 2013;71(1):1-2.
22. Brittain K, Mellins CA, Remien RH, et al. Impact of HIV-status disclosure on HIV viral load in pregnant and postpartum women on antiretroviral therapy. *J Acquir Immune Defic Syndr* 2019;81(4):379-86.
23. Trafford Z, Gomba Y, Colvin CJ, Iyun VO, Phillips TK, Brittain K, Myer L, Abrams EJ, Zerbe A. Experiences of HIV-positive postpartum women and health workers involved

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

with community-based antiretroviral therapy adherence clubs in Cape Town, South Africa. *BMC Public Health* 2018 Dec;18(1):1-3.

24. Manjate Cuco RM, Munguambe K, Bique Osman N, et al. Male partners' involvement in prevention of mother-to-child HIV transmission in sub-Saharan Africa: a systematic review. *SAHARA-J* 2015;12(1):87-105.
25. Chanyalew H, Girma E, Birhane T, et al. Male partner involvement in HIV testing and counseling among partners of pregnant women in the Delanta District, Ethiopia. *Plos One* 2021;16(3):e0248436.
26. Dinh TH, Delaney KP, Goga A, Jackson D, Lombard C, Woldesenbet S, Mogashoa M, Pillay Y, Shaffer N. Impact of maternal HIV seroconversion during pregnancy on early mother to child transmission of HIV (MTCT) measured at 4-8 weeks postpartum in South Africa 2011-2012: a national population-based evaluation. *PloS One* 2015;10(5):e0125525.
27. Moher D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;4(1):1-9.
28. Jackson R, Ameratunga S, Broad J, Connor J, Lethaby A, Robb G, et al. The GATE frame: critical appraisal with pictures. *Evid Based Med* 2006;11(2):35-8.
29. Ouzzani M, Hammady H, Fedorowicz Z, Elmagarmid A. Rayyan-a web and mobile app for systematic reviews. *Syst Rev* 2016;5:210
30. Higgins JP, Altman DG, Gotzsche PC, et al. The Cochrane Collaboration's tool for assessing risk of bias in randomised trials. *BMJ* 2011;343:1–9. https://doi.org/10.1210/jcem-43-6-1203.
31. Sterne JAC, Hernán MA, Reeves BC, et al. ROBINS-I: a tool for assessing risk of bias in non-randomized studies of interventions. *BMJ* 2016;355;i4919; doi: 10.1136/bmj.i4919.
32. Hayden JA, Coté P, Bombardier C. Evaluation of the Quality of Prognosis Studies in Systematic Reviews. *Ann Intern Med* 2006;144(6):427-438
33. Higgins JP, Thompson SG, Deeks JJet al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557.
34. StataCorp. 2019. Stata Statistical Software: Release 16. College Station, TX: StataCorp LLC.
35. Akers J, Aguiar-Ibáñez R, Baba-Akbari A. Systematic reviews: CRD's guidance for undertaking reviews in health care. York, UK: Centre for Reviews and Dissemination, University of York 2013.
36. Iorio A, Spencer FA, Falavigna M, et al. Use of GRADE for assessment of evidence about prognosis: rating confidence in estimates of event rates in broad categories of patients. *BMJ* 2015;350.

For peer review only

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

37. Foroutan F, Guyatt G, Zuk V, et al. GRADE Guidelines 28: Use of GRADE for the assessment of evidence about prognostic factors: rating certainty in identification of groups of patients with different absolute risks. *J Clinl Epidemiol* 2020;121:62-70.

For peer review only

Figures

Figure 1: A summary of reported and potential role of male-partner HIV disclosure within the PMTCT causal pathways.

Figure 2: Flowchart showing detailed summary of the search and screening processes

Supplementary material

Supplementary file 1:

Title: PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol

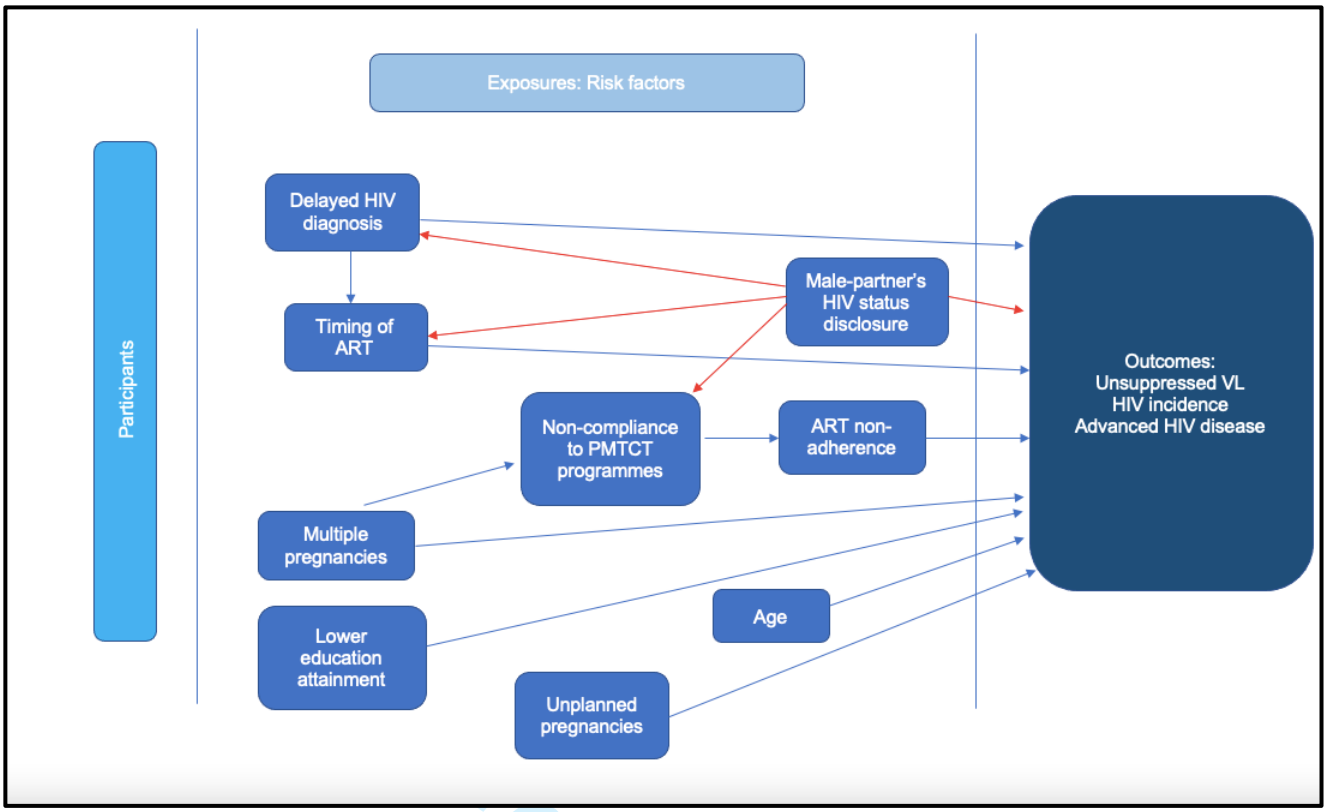
Format: PDF

Supplementary file 2:

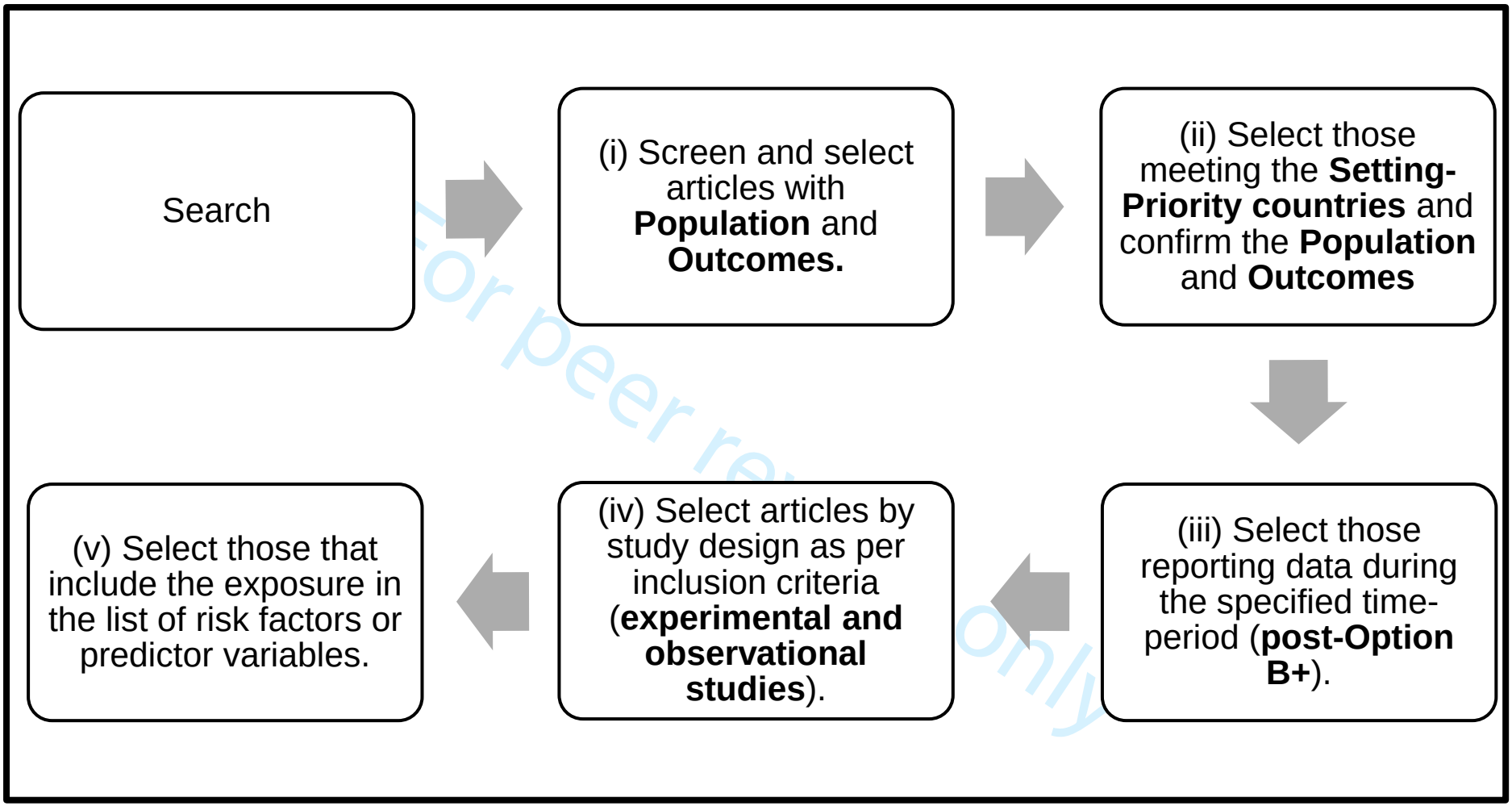
Title: Preliminary search strategy generated to date from PubMed, SCOPUS, Web of Science and Cochrane Library.

Format: PDF

For peer review only



For peer review only



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

PRISMA-P (Preferred Reporting Items for Systematic review and Meta-Analysis Protocols) 2015 checklist: recommended items to address in a systematic review protocol*

Section and topic	Item No	Checklist item	Page no.
ADMINISTRATIVE INFORMATION			
Title:			
Identification	1a	Identify the report as a protocol of a systematic review	6
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	N/A
Registration	2	If registered, provide the name of the registry (such as PROSPERO) and registration number	6
Authors:			
Contact	3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	10
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	N/A
Support:			
Sources	5a	Indicate sources of financial or other support for the review	10
Sponsor	5b	Provide name for the review funder and/or sponsor	N/A
Role of sponsor or funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	N/A
INTRODUCTION			
Rationale	6	Describe the rationale for the review in the context of what is already known	3-5
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	7
METHODS			
Eligibility criteria	8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	6-7
Information sources	9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	7
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	8
Study records:			
Data	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7-8

For peer review only

management			
Selection process	11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	8
Data collection process	11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	8
Data items	12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	8
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	8
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	9
Data synthesis	15a	Describe criteria under which study data will be quantitatively synthesised	9
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I^2, Kendall's τ)	9
	15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	9
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	9
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	8
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	9

*** It is strongly recommended that this checklist be read in conjunction with the PRISMA-P Explanation and Elaboration (cite when available) for important clarification on the items. Amendments to a review protocol should be tracked and dated. The copyright for PRISMA-P (including checklist) is held by the PRISMA-P Group and is distributed under a Creative Commons Attribution Licence 4.0.**

From: Shamseer L, Moher D, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart L, PRISMA-P Group. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. BMJ. 2015 Jan 2;349(jan02 1):g7647.

For peer review only

PUBMED

Search	Query
#8	Search: ((((#1) AND (#2 OR #3)) AND (#4)) AND (#5)) AND (#6) Filters: from 2011/1/1 - 2021/9/15
#7	Search: ((((#1) AND (#2 OR #3)) AND (#4)) AND (#5)) AND (#6)
#6	Search: angola OR botswana OR burundi OR cameroon OR chad OR "cote d'ivoire" OR "democratic republic of congo" OR ethiopia OR ghana OR "ivory coast" OR kenya OR lesotho OR malawi OR mozambique OR namibia OR nigeria OR eswatini OR swaziland OR "south africa" OR tanzania OR uganda OR zambia OR zimbabwe
#5	Search: disclos* OR "self disclosure" OR self-disclosure OR non-disclosure
#4	Search: spous* OR "sexual partner*" OR partner* OR male* OR husband* OR man OR men
#3	Search: "option B+" OR "mother-to-child transmission" OR "maternal-to-child transmission" OR "maternal-fetal infection transmission" OR mtct OR pmtct
#2	Search: matern* OR wom* OR pregnan* OR gestation* OR intrauterine OR obstetric OR delivery OR childbirth OR postpart* OR post-part* OR "breast feeding" OR breastfeeding OR lactat* OR parturition OR peripartum OR ante-natal OR antenatal OR anc OR pre-natal OR prenatal OR newborn OR neonat* OR bab*
#1	Search: "HIV infection" OR "HIV infections" OR HIV OR HIV1 OR HIV2 OR "human immuno-deficiency virus" OR "human immunodeficiency virus" OR "acquired immunodeficiency syndrome" OR "acquired immuno-deficiency syndrome" OR "acquired immune-deficiency syndrome" OR AIDS

SCOPUS

(angola OR botswana OR burundi OR cameroon OR chad OR "cote d'ivoire" OR "democratic republic of congo" OR ethiopia OR ghana OR "ivory coast" OR kenya OR lesotho OR malawi OR mozambique OR namibia OR nigeria OR eswatini OR swaziland OR "south africa" OR tanzania OR uganda OR zambia OR zimbabwe) AND TITLE-ABS-KEY (disclos* OR "self disclosure" OR self-disclosure OR non-disclosure) AND TITLE-ABS-KEY (spous* OR "sexual partner*" OR partner* OR male* OR husband* OR man OR men) AND TITLE-ABS-KEY ("option b+" OR "mother-to-child transmission" OR "maternal-to-child transmission" OR "maternal-fetal infection transmission" OR mtct OR pmtct OR matern* OR wom* OR pregnan* OR gestation* OR intrauterine OR obstetric OR delivery OR childbirth OR postpart* OR post-part* OR "breast feeding" OR breastfeeding OR lactat* OR parturition OR peripartum OR ante-natal OR antenatal OR anc OR pre-natal OR prenatal OR newborn OR neonat* OR bab*) AND TITLE-ABS-KEY ("hiv infection" OR "hiv infections" OR hiv OR hiv1 OR hiv2 OR "human immuno-deficiency virus" OR "human immunodeficiency virus" OR "acquired immunodeficiency syndrome" OR "acquired immuno-deficiency syndrome" OR "acquired immune-deficiency syndrome" OR aids)) AND PUBYEAR > 2010 AND PUBYEAR < 2022

WEB OF SCIENCE

"HIV infection" OR "HIV infections" OR HIV OR HIV1 OR hive OR "human immuno-deficiency virus" OR "human immunodeficiency virus" OR "acquired immunodeficiency syndrome" OR "acquired immuno-deficiency syndrome" OR "acquired immune-deficiency syndrome" OR AIDS (All Fields) and "option B+" OR "mother-to-child transmission" OR "maternal-to-child transmission" OR "maternal-fetal infection transmission"

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

OR mtct OR pmtct OR matern* OR wom* OR pregnan* OR gestation* OR intrauterine OR obstetric OR delivery OR childbirth OR postpart* OR post-part* OR "breast feeding" OR breastfeeding OR lactat* OR parturition OR peripartum OR ante-natal OR antenatal OR anc OR pre-natal OR prenatal OR newborn OR neonat* OR bab* (All Fields) and spous* OR "sexual partner*" OR partner* OR male* OR husband* OR man OR men (All Fields) and disclos* OR "self disclosure" OR self-disclosure OR non-disclosure (All Fields) and angola OR botswana OR burundi OR cameroon OR chad OR "cote d'ivoire" OR "democratic republic of congo" OR ethiopia OR ghana OR "ivory coast" OR kenya OR lesotho OR malawi OR mozambique OR namibia OR nigeria OR eswatini OR swaziland OR "south africa" OR tanzania OR uganda OR zambia OR zimbabwe (All Fields) | 786 results

Timespan: 2011-01-01 to 2021-09-15 (Publication Date)

COCHRANE LIBRARY

"HIV infection" OR "HIV infections" OR HIV OR HIV1 OR HIV2 OR "human immuno-deficiency virus" OR "human immunodeficiency virus" OR "acquired immunodeficiency syndrome" OR "acquired immuno-deficiency syndrome" OR "acquired immune-deficiency syndrome" OR AIDS in Title Abstract Keyword AND "option B+" OR "mother-to-child transmission" OR "maternal-to-child transmission" OR "maternal-fetal infection transmission" OR mtct OR pmtct OR matern* OR wom* OR pregnan* OR gestation* OR intrauterine OR obstetric OR delivery OR childbirth OR postpart* OR post-part* OR "breast feeding" OR breastfeeding OR lactat* OR parturition OR peripartum OR ante-natal OR antenatal OR anc OR pre-natal OR prenatal OR newborn OR neonat* OR bab* in Title Abstract Keyword AND spous* OR "sexual partner*" OR partner* OR male* OR husband* OR man OR men in Title Abstract Keyword AND disclos* OR "self disclosure" OR self-disclosure OR non-disclosure in Title Abstract Keyword AND angola OR botswana OR burundi OR cameroon OR chad OR "cote d'ivoire" OR "democratic republic of congo" OR ethiopia OR ghana OR "ivory coast" OR kenya OR lesotho OR malawi OR mozambique OR namibia OR nigeria OR eswatini OR swaziland OR "south africa" OR tanzania OR uganda OR zambia OR zimbabwe in Title Abstract Keyword - (Word variations have been searched)

Timespan: 2011-01-01 to 2021-09-15

For peer review only